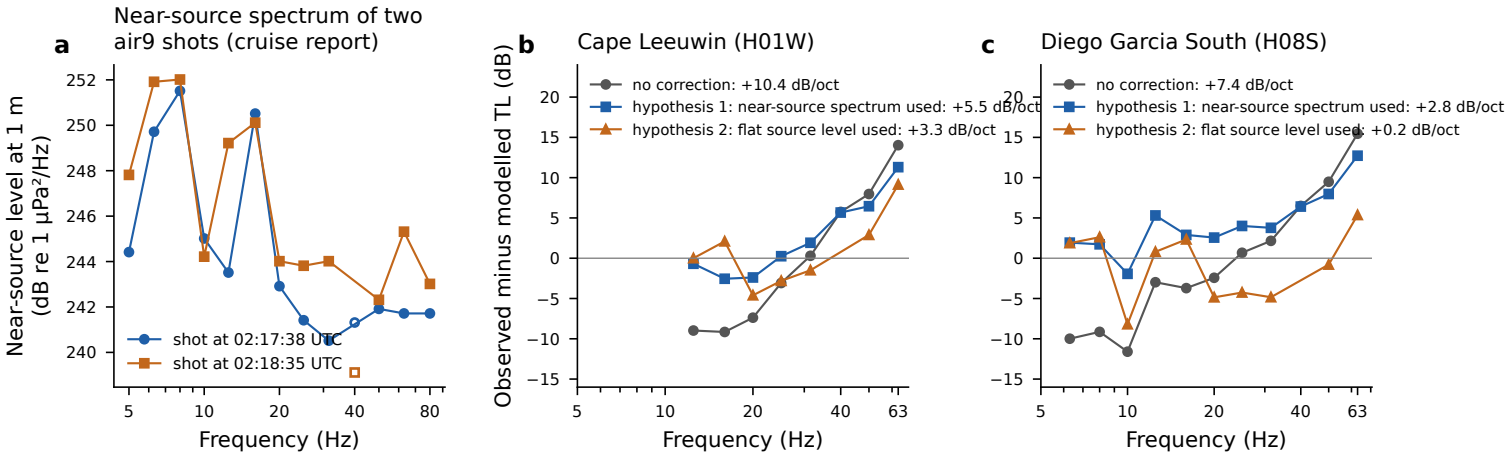
The near-source reference removes part of the airgun slope, not all of it



Plain-language note: The 2001 cruise report gives the depth of the airguns and a calibrated recording near the guns. We test two ideas about the source level that the 2004 report used to calculate sound loss. If the correction were correct, each line in (b) and (c) would be flat. With either idea, the Cape Leeuwin line still rises too much (limit 3 dB per octave), so neither idea is supported.

Technical note: EW0112 cruise report pp. 5-7 (tow depth 11-12 m; Ball hydrophone calibrated, about 30 m ahead of the array, depth not stated); Fig. 2 DIGITISED (200-dpi scan; 1/3-octave means; open markers below 5 % of full scale; scaling to 1 m from the '@ 1M'/'@ receiver' peak ratio, 24.7-26.8 m; 1-m peak 232.6-233.0 dB re 1 µPa, as Blackman 2004 '230-240 dB'). Residual r = Fig. 23 observed TL − module KRAKEN TL (model M1, z_s 10 m; audit air9_residuals.csv). Hypothesis 1: r = near-field reference factor 10 log10(<|e^{ikr1}/r1 − e^{ikr2}/r2|²> R1m²), point source, Ball at x 30 m, z 15 m (nominal). Hypothesis 2: r = C − S0, S0 = Ball spectrum minus that factor; level not tested. Pre-registered (air9_ew0112.py, ecbca309): over 45 geometries (z_s 9/10/12, x 20/30/40, z 5-30 m) both stations pass slope in 0 % (1) and 31 % (2) of cases, limit 50 %: NOT SUPPORTED. 20-gun proxy array is worse. Not a likelihood.